Mathematics for Computer Science – CRT

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Table of contents

导引.

数论部分，主要目标是解方程。群论部分，主要目标是群结构。

思路.

- 数论部分，目前我们学习解方程方法包括：一元一次同余方程，一元 n 次同余方程。接下来需要扩展到一元同余方程组的解法。未来是多元同余方程等等。
- 群论部分，主要目标是群结构。群结构与解同余方程有什么关系吗？

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目标

给出一种求解一元一次同余方程组的高效算法；探讨群结构与解同余方程组的关系。

导引.

Chinese Remainder Theorem（中国剩余定理），或称为*中国余数定理*则更准确。

历史.

中国剩余定理有着满满的中国传统文化元素，与之相关的名人包括：春秋时期的孙子、秦汉时期的韩信、宋朝的沈括和秦久韶等。南北朝时期（公元 5 世纪）的数学著作《孙子算经》卷下第二十六题所谓“物不知数”问题则为国人所熟知：有物不知其数，三三数之剩二，五五数之剩三，七七数之剩二。问物几何？

Motivation.

中国剩余定理讨论一元同余方程组的高效解法。

Example (Example 1.)

Suppose we wish to solve:

$$x \equiv 2 \pmod{5}$$
$$x \equiv 3 \pmod{7}$$

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Suppose we wish to solve:

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Solution.

1. We have $x = 5t + 2$ from the first congruence, for $t \in \mathbb{Z}$;

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2. Substitute for x in the second congruence, $5t + 2 \equiv 3 \pmod{7}$;

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1. We have $x = 5t + 2$ from the first congruence, for $t \in \mathbb{Z}$;
2. Substitute for x in the second congruence, $5t + 2 \equiv 3 \pmod{7}$;
3. Simplifying, we get $5t \equiv 1 \pmod{7}$;

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2. Substitute for x in the second congruence, $5t + 2 \equiv 3 \pmod{7}$;
3. Simplifying, we get $5t \equiv 1 \pmod{7}$;
4. Multiplying both sides by 5^{-1}, we get $t = 7s + 3$ for $s \in \mathbb{Z}$;

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2. Substitute for x in the second congruence, $5t + 2 \equiv 3 \pmod 7$;
3. Simplifying, we get $5t \equiv 1 \pmod 7$;
4. Multiplying both sides by 5^{-1}, we get $t = 7s + 3$ for $s \in \mathbb{Z}$;
5. Finally, $x = 35s + 17$, which means $x \equiv 17 \pmod{35}$.

Question.

思考.

以上求解一元同余方程组的方法是否高效？如果不高效，能否给出更高效的算法？

The Chinese Remainder Theorem–CRT.

For any system of equations like this, the *Chinese Remainder Theorem*, short for CRT, tells us there is always a unique solution up to a certain modulus, and describes how to find the solution efficiently.

Theorem

Let p, q be primes, $n = pq$. For each $a \in \mathbb{Z}_p$, $b \in \mathbb{Z}_q$, there is a unique x, $0 \leq x < n$ such that $x \equiv a \pmod{p}$ and $x \equiv b \pmod{q}$.

The Chinese Remainder Theorem–CRT.

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Let p, q be two relatively prime positive integers, $n = pq$. For each $a \in \mathbb{Z}_p$, $b \in \mathbb{Z}_q$, there is a unique x, $0 \leq x < n$ such that $x \equiv a \pmod{p}$ and $x \equiv b \pmod{q}$.

Proof Idea

1. Given a and p, how can we find some c s.t. $ac \equiv a \pmod{p}$?

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1. Given a and p, how can we find some c s.t. $ac \equiv a \pmod{p}$?
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3. 模 p 下为 1 的数可能有哪些（可参考 CINTA 附录 D.）？

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4. Find some c s.t. $acc^{-1} \equiv a \pmod{p}$

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2. c must be some 1 under modulo p
3. 模 p 下为 1 的数可能有哪些（可参考 CINTA 附录 D.）？
4. Find some c s.t. $acc^{-1} \equiv a \pmod{p}$
5. Then x must be something like that $x = (acc^{-1} + bdd^{-1})$, what should be c and d?

The Chinese Remainder Theorem–CRT.

Theorem

Let p, q be coprime positive integers, $n = pq$. For each $a \in \mathbb{Z}_p$, $b \in \mathbb{Z}_q$, there is a unique x, $0 \leq x < n$ such that $x \equiv a \pmod p$ and $x \equiv b \pmod q$.

Proof.

By construction. Since p, q are coprime, there exist p_1 and q_1 such that $p_1 \equiv p^{-1} \pmod q$ and $q_1 \equiv q^{-1} \pmod p$. Define x by:

$$x = aqq_1 + bpp_1$$

It is easy to check that x satisfies both equations. It remains to show no other solutions exist modulo n. Suppose $\exists z \neq x$ is another solution. Then $(z - x) = tp$ and $(z - x) = sq$, for some $t, s \in \mathbb{Z}$. Since p and q are coprime, then $(z - x) = kpq$, for $k \in \mathbb{Z}$. Hence $z \equiv x \pmod n$.

$\square$

Example.

Example (Example 2.)

Suppose we wish to solve:

$$x \equiv 2 \pmod{5}$$
$$x \equiv 3 \pmod{7}$$

Solution.

1. Let $a = 2$, $b = 3$, $p = 5$, $q = 7$, $n = pq = 35$;
2. Compute $p_1 \equiv p^{-1} \pmod{q}$ and $q_1 \equiv q^{-1} \pmod{p}$ using egcd algorithm; $p_1 = 3$, $q_1 = 3$;
3. $x \equiv aqq_1 + bpp_1 \pmod{n}$; $x \equiv 17 \pmod{35}$;
4. It is easy to check that x is a correct solution.

Exercise.

求解以下方程.

Suppose we wish to solve:

$$
\begin{aligned}
x &\equiv 3 \pmod{11} \\
x &\equiv 4 \pmod{13}
\end{aligned}
$$

Exercise.

求解以下方程.

Suppose we wish to solve:

$$\begin{aligned} x &\equiv 3 \pmod{11} \\ x &\equiv 4 \pmod{13} \end{aligned}$$

Ans.

69

Generalization.

For several equations, we have a generalized version of CRT.

Theorem

Let $m_1, m_2, \ldots, m_n$ be a set of pairwise relatively prime positive integers. Then the system of n equations:

$$x \equiv a_1 \pmod{m_1}$$
$$\ldots$$
$$x \equiv a_n \pmod{m_n}$$

has a unique solution for x modulo M where $M = m_1 m_2 \ldots m_n$.

Generalization.

Proof.

By construction. Let $M = \prod_{i=1}^{n} m_i$, $b_i = M/m_i$, $b'_i = b_i^{-1} \pmod{m_i}$. Then

$$y = \sum_{i=1}^{n} a_i b_i b'_i \pmod{M}$$

is the unique solution. $\square$

思考题-1.

思考题.

设 m 和 n 为互素的正整数，$a > 0$ 为一个正整数，如果

$$x \equiv a \pmod{m}$$
$$x \equiv a \pmod{n}$$

x 模 mn 等于什么？为什么？

思考题-2.

如果同余方程组的模数 $m_1, m_2, \ldots, m_n$ 不两两互质，那么上述方程组是否还有解？如果有解，怎么求？请思考下面这题。

思考题.

判定下列两个同余方程组是否有解。若有，求出模适当模数下的所有解。

(a) $\begin{cases} x \equiv 7 \pmod{12} \\ x \equiv 3 \pmod{8} \end{cases}$　　(b) $\begin{cases} x \equiv 5 \pmod{12} \\ x \equiv 2 \pmod{8} \end{cases}$

思考题-2.

如果同余方程组的模数 $m_1, m_2, \ldots, m_n$ 不两两互质，那么上述方程组是否还有解？如果有解，怎么求？请思考下面这题。

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(a) $\begin{cases} x \equiv 7 \pmod{12} \\ x \equiv 3 \pmod{8} \end{cases}$

(b) $\begin{cases} x \equiv 5 \pmod{12} \\ x \equiv 2 \pmod{8} \end{cases}$

思路

模数不互素，也就是说有大于 1 的公共因子 d，这个 d 能起到什么作用？其次，反向应用思考题-1 的结论，从一个方程式化为多个方程式。

A perspective from Abstract Algebra.

提醒：本节关注的不是算法，而是 CRT 背后的思想。

Motivation.

Let $n = pq$, where $p, q > 1$ are relatively prime. Given a positive integer x, it can be expressed as a unique pair $\big([x \bmod p], [x \bmod q]\big)$.

A perspective from Abstract Algebra.

Theorem

Let $p, q > 1$ be coprime, $n = pq$. Then

$$\mathbb{Z}_n \cong \mathbb{Z}_p \times \mathbb{Z}_q \quad (\textit{as an additive group})$$

and,

$$\mathbb{Z}_n^* \cong \mathbb{Z}_p^* \times \mathbb{Z}_q^* \quad (\textit{as a multiplicative group}).$$

Proof.

1. Define $\phi : \mathbb{Z}_n \to \mathbb{Z}_p \times \mathbb{Z}_q$ by:

$$\phi(x) \triangleq \big([x \bmod p], [x \bmod q]\big)$$

2. Show ϕ is bijective.
3. Check that $\phi(x)$ preserves the group operation.

A perspective from Abstract Algebra.

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2. Show ϕ is bijective.
3. Check that $\phi(x)$ preserves the group operation.

The proof that it is an isomorphism from $\mathbb{Z}_n^*$ to $\mathbb{Z}_p^* \times \mathbb{Z}_q^*$ is similar. □

Remark.

Remark

首先，注意到 $\mathbb{Z}_p \times \mathbb{Z}_q$ 是加法群，请验证！

How to prove the bijection?

Proof.

证明映射 ϕ 是一种双射，即证明 ϕ 是满射且单射。满射显然，因为根据中国剩余定理，任意序对中的两个同余式在模 n 下存在唯一解。证明单射即证明，如果对任意整数 $0 \leq a, b < n$，有 $\left([a \bmod p], [a \bmod q]\right) = \left([b \bmod p], [b \bmod q]\right)$，则 $a = b$。再次根据中国剩余定理可得。 □

How to prove the isomorphism?

Proof.

证明映射 ϕ 保持群操作，即需要证明：

$$\begin{aligned}\phi(a+b) &= \big([(a+b) \bmod p], [(a+b) \bmod q]\big) \\ &= \big([a \bmod p], [a \bmod q]\big) + \big([b \bmod p], [b \bmod q]\big) \\ &= \phi(a) + \phi(b)\end{aligned}$$

□

Example.

Example (Example 3.)

Take $n = 15 = 5 \cdot 3$. $\mathbb{Z}_n^* = \{1, 2, 4, 7, 8, 11, 13, 14\}$ is isomorphic to $\mathbb{Z}_5^* \times \mathbb{Z}_3^*$ since we can give the following correspondence:

$$1 \leftrightarrow (1, 1) \quad 2 \leftrightarrow (2, 2) \quad 4 \leftrightarrow (4, 1) \quad 7 \leftrightarrow (2, 1)$$

$$8 \leftrightarrow (3, 2) \quad 11 \leftrightarrow (1, 2) \quad 13 \leftrightarrow (3, 1) \quad 14 \leftrightarrow (4, 2)$$

Example.

Example (Example 4.)

To compute $14 \cdot 13 \bmod 15$. Since $14 \leftrightarrow (4, 2)$ and $13 \leftrightarrow (3, 1)$, we have:

$$(4, 2) \cdot (3, 1) = ([4 \cdot 3 \bmod 5], [2 \cdot 1 \bmod 3]) = (2, 2).$$

Note that $(2, 2) \leftrightarrow 2$, which is the correct answer.

Example.

Example (Example 5.)

To compute $11^{53} \bmod 15$. Since $11 \leftrightarrow (1,2)$ and $2 \equiv -1 \bmod 3$ we have:

$$(1,2)^{53} = ([1^{53} \bmod 5], [(-1)^{53} \bmod 3]) = (1, -1 \bmod 3) = (1,2).$$

Thus, $11^{53} \bmod 15 = 11$

Example.

Example (Example 6.)

设 $n = p \cdot q$ 为合数，p 和 q 是两个不同的奇素数。以下等式在模 n 的意义上有多少个解？

$$x^2 \equiv 1 \pmod{n}$$

为此，需要解以下两个同余方程

$$x^2 \equiv 1 \pmod{p}$$
$$x^2 \equiv 1 \pmod{q}$$

以上两式分别有两个不同的解。因此在模 n 的意义上总共有 4 个解。同时，这也就解决了上一章的一个遗留问题。

Example.

Example (Example 7.)

To compute $12^{12} \bmod 143$

Example.

Example (Example 7.)

To compute $12^{12} \bmod 143$ 解:

- 记 $n = 143$, $n = p \cdot q = 13 \cdot 11$;
- 求: $12^{12} \bmod 13$ 和 $12^{12} \bmod 11$;
- 所以, $12^{12} \bmod 143 = 1$.

Little thought.

Remark

A practical application: if we have many computations to perform on $x \in \mathbb{Z}_n^*$ *(e.g. RSA signing and decryption), we can convert* x *to* $(a, b) \in \mathbb{Z}_p^* \times \mathbb{Z}_q^*$ *and do all the computations on* a *and* b *instead before converting back.*
This is often cheaper because for many algorithms, doubling the size of the input more than doubles the running time.

Homework.

Homework.

1. Use CRT to solve:

$$\begin{aligned} x &\equiv 8 \pmod{11} \\ x &\equiv 13 \pmod{19} \end{aligned}$$

2. Use CRT to solve the system of congruences:

$$\begin{aligned} x &\equiv 1 \pmod{5} \\ x &\equiv 2 \pmod{7} \\ x &\equiv 3 \pmod{9} \\ x &\equiv 4 \pmod{11} \end{aligned}$$

3. Write a program (in C or Python) to solve CRT.

Homework.

Homework.

4. Complete the proof that it is an isomorphism from $\mathbb{Z}_n^*$ to $\mathbb{Z}_p^* \times \mathbb{Z}_q^*$.
5. Let $p = 5$ and $q = 7$, $n = p \cdot q$. Please explicitly give the correspondence between $\mathbb{Z}_n^*$ and $\mathbb{Z}_p^* \times \mathbb{Z}_q^*$. Hint: Programming is permitted.

Homework.

Homework.

6. 求解 $x^{113} \equiv 15 \pmod{221}$[提示：这是一道似曾相识的习题，然而中国剩余定理能保证你顺利完成目标。]